

Module 4 – AI & Deployment

1. Deep Learning

What is Deep Learning?

Deep learning is a subset of machine learning that uses neural networks to learn complex patterns from large datasets.

Key Features

- Uses multiple layers (deep networks)
 - Handles complex data like images and text
 - Powers modern AI systems
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2. TensorFlow

What is TensorFlow?

TensorFlow is an open-source library used for building and training deep learning models.

Why Use TensorFlow

- Scalable and powerful
- Supports neural networks
- Widely used in industry

Example:

```
import tensorflow as tf

model = tf.keras.Sequential()
```

3. Generative AI

What is Generative AI?

Generative AI is a type of AI that can create new content such as text, images, and code.

Examples

- Chatbots
 - Image generators
 - Text generation tools
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4. Model Deployment

What is Deployment?

Deployment is the process of making a trained model available for real-world use.

Methods

- Web APIs
 - Cloud platforms
 - Mobile apps
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5. Best Practices

Important Guidelines

- Use clean and structured code
 - Validate models before deployment
 - Monitor performance
 - Ensure scalability
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6. Key Concepts to Remember

- Deep learning uses neural networks
 - TensorFlow builds AI models
 - Generative AI creates content
 - Deployment makes models usable
 - Best practices ensure quality
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7. Summary

AI and deployment are the final steps in data science. By using deep learning, TensorFlow, and deployment techniques, you can build intelligent systems and bring them into real-world applications.